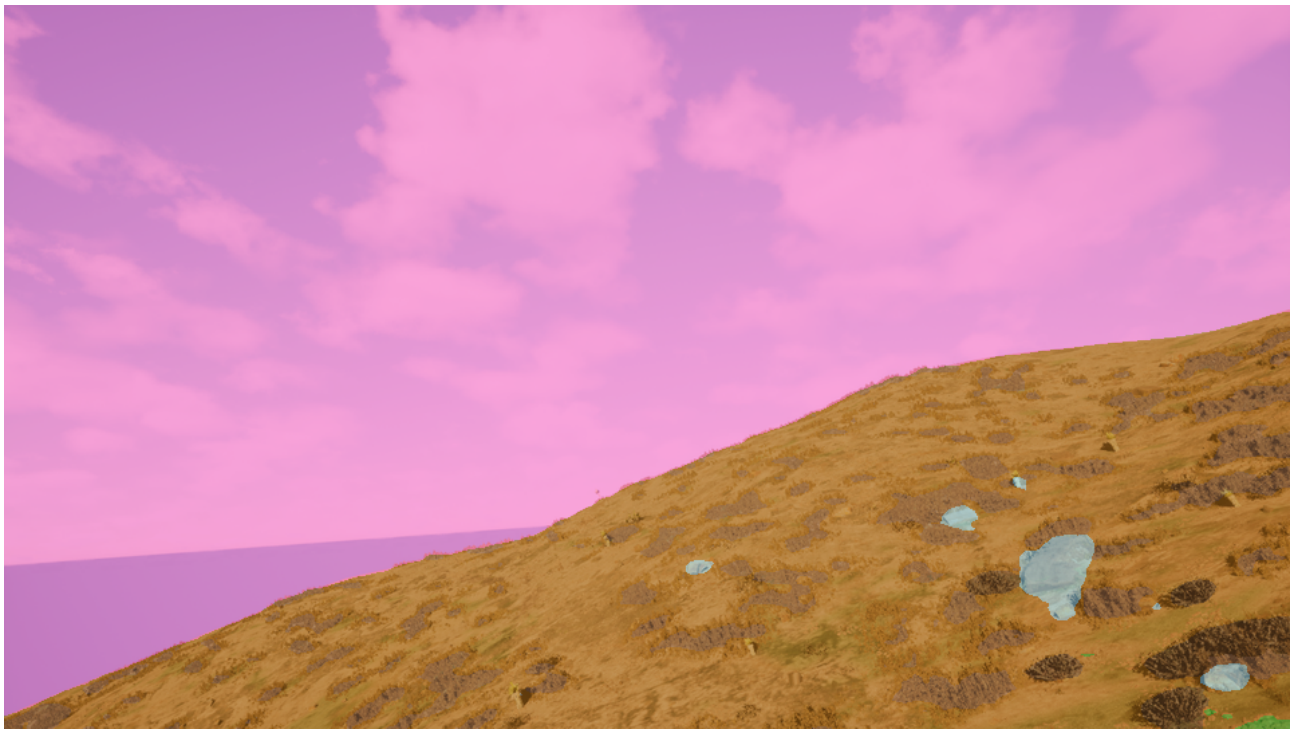
TERRAIN INTELLIGENCE BRIEFING

EXERCISE USE ONLY — NOT FOR OPERATIONAL USE

Session ID	492b8abd
Generated	2026-04-17 18:05:58 UTC
Source File	cc0000016.png
Model	SegFormer-B2 | mIoU: 0.6109 | Flowers IoU: 0.9801 | Logs IoU: 0.6504
Classes	10 defined | 8 present in dataset | 2 absent (Trees, Water)

Terrain Analysis



18.1 **LOW Terrain heavily obstructed. Do not advance.**

PRIMARY ACTION: RETREAT

Traversability score is 18.1/100 (LOW), indicating terrain heavily obstructed. do not advance. Threat assessment is HIGH: Active alerts: Logs, Flowers. 5 zone(s) blocked.

Terrain Composition

Class	Coverage %	Traversable	Status
Gravel	0.00%	Yes	Active
Rocks	5.13%	No	Active
Dirt	1.10%	Yes	Active
Grass	0.06%	Yes	Active
Sky	0.71%	No	Active
Logs	26.46%	No	ALERT
Flowers	66.54%	No	ALERT

* Trees (class 5) and Water (class 6) are excluded — zero pixels across the entire FalconCloud synthetic training dataset. The model was not trained on these classes and cannot predict them.

Threat Assessment

THREAT LEVEL: HIGH

Active Alerts

Logs — 26.46% coverage OBSTACLE — potential cover or concealment detected

Flowers — 66.54% coverage ANOMALY — unusual vegetation pattern detected

Blocked / Avoid Zones

Row 0 Col 0 (score: 20.0)
Row 0 Col 1 (score: 20.0)
Row 0 Col 2 (score: 20.0)
Row 1 Col 0 (score: 20.0)
Row 1 Col 1 (score: 19.0)
Row 1 Col 2 (score: 15.5)
Row 2 Col 0 (score: 16.5)
Row 2 Col 1 (score: 12.0)
Row 2 Col 2 (score: 19.7)

Technical Details

Parameter	Value
Architecture	SegFormer-B2
Encoder	Mix Transformer (MiT-B2)
Decoder	All-MLP Head
Input Resolution	512 × 512
Output Classes	10 (8 present in dataset)
Parameters	~25M
Training Loss	CrossEntropy + Dice (weighted)
Optimizer	AdamW + warmup scheduler
Hard Mining	Qdrant vector store — 638 hard examples

Per-Class IoU — Best Checkpoint (run_best.pth)

Class ID	Class Name	IoU	Notes
0	Sand	0.4821	
1	Gravel	0.5103	
2	Rocks	0.5892	
3	Dirt	0.6217	
4	Grass	0.5934	
5	Trees	*N/A*	Not in training dataset
6	Water	*N/A*	Not in training dataset
7	Sky	0.7841	
8	Logs	0.6504	Rare class ★
9	Flowers	0.9801	Rare class ★

Training Summary

Metric	Value
Total Epochs	22
Train Images	2857
Val Images	317
Hard Examples Mined	638 (Qdrant)
Best mIoU	0.6109

Best Flowers IoU	0.9801
Best Logs IoU	0.6504
Dataset Source	FalconCloud synthetic — 8 of 10 classes present